Decent Coaching Classes

X Semi

Academic Conquest - VII

Subject: Science and Technology (Ch. 1 Gravitation) **Answer Key | Marks: 30**

Q.1 Choose the correct alternative. (1 mark each) **[5]**

1. (b) Newton
2. (b) 9.8 m/s²
3. (c) an ellipse
4. (c) Newton
5. (a) 11.2 km/s

Q.2 One-sentence answers (any 3) (1 mark each) **[3]**

(a) The line joining the planet and the Sun sweeps equal areas in equal intervals of time.

(b) When an object moves under the influence of the force of gravity alone, it is said to be in free fall.

(c) The force acting on an object moving along a circle, directed towards the centre of the circle, is called centripetal force.

(d) Henry Cavendish.

Q.3 Scientific reasons (any 2) (2 marks each) **[4]**

(a) Weight W = mg. The value of g on the moon is about 1/6th of its value on the earth, while the mass is the same. Hence the weight on the moon is about 1/6th of that on the earth.

(b) In air, friction and buoyant force act on a falling object along with gravity, so it is not under the influence of gravity alone. In vacuum there is no air, so only gravity acts. (E.g. a feather and a stone reach the ground together only in vacuum.)

(c) The earth's gravitational force constantly pulls the stone downwards. It reduces the velocity of the stone until it becomes zero, after which the stone starts moving down towards the centre of the earth.

Q.4 Answer in brief (any 3) (2 marks each) **[6]**

(a) First law: The orbit of a planet is an ellipse with the Sun at one of the foci. (1) Third law: The square of the period of revolution of a planet around the Sun is directly proportional to the cube of its mean distance from the Sun, $T^2 \propto r^3$. (1)

(b) Any two points (1 mark each):

Mass: amount of matter in an object; SI unit kg; scalar quantity; same everywhere.

Weight: force with which the earth attracts an object (W = mg); SI unit Newton; vector quantity; changes from place to place.

(c) $g = GM/R^2$, where G = universal gravitational constant, M = mass of the earth, R = radius of the earth (1 mark). Value of g on the surface of the earth = 9.8 m/s² (1 mark).

(d) Escape velocity is the minimum initial velocity with which an object must be thrown upwards so that it escapes the gravitational pull of the earth and does not fall back (1 mark). For the earth it is about 11.2 km/s (1 mark).

Q.5 Answer the following (any 2) (3 marks each) **[6]**

(a) Activity: Tie a stone to one end of a string and whirl it in a circle, holding the other end in hand (1). The string pulls the stone towards the centre of the circle. This force, directed towards the centre on an object moving along a circle, is the centripetal force (1). If the string is released, the force stops and the stone flies off along the tangent to the circle. Magnitude: $F = mv^2/r$ (1).

(b) Statement (2): Every object in the universe attracts every other object with a force which is directly proportional to the product of their masses and inversely proportional to the square of the distance between them. Formula (1): $F = G\, m_1 m_2 / d^2$, where G is the universal gravitational constant (6.673×10^{-11} N m² kg^{-2}).

(c) $g = GM/r^2$. As we go above the surface of the earth, the distance r from the centre increases, so g decreases (1). For heights small compared with the earth's radius (e.g. an aeroplane 10 km high) the

change is negligible (1). For large heights the decrease is large, e.g. $g = 9.8$ m/s² at the surface, 8.7 m/s² at 400 km and 0.225 m/s² at 35700 km (1).

Q.6 Numericals (any 2) (3 marks each) **[6]**

(a) $m = 60$ kg. Weight on earth $= mg = 60 \times 10 = 600$ N (1½). Weight on moon $= 60 \times 1.6 = 96$ N (1½).

(b) $u = 0$, $t = 3$ s, $a = g = 10$ m/s². $v = u + gt = 0 + 10 \times 3 = 30$ m/s (1½). $s = ½ gt^2 = ½ \times 10 \times 3^2 = 45$ m (1½).

(c) $F = G m_1 m_2 / d^2$ (1) $= (6.67 \times 10^{-11} \times 10 \times 20) / 1^2$ (1) $= 1.334 \times 10^{-8}$ N (1).